

G-05-PHASE-CLOSURE v1: G-concentration closed, bottleneck E-weak -¿ E-strong

ZFL Lab

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PHASE G CLOSED: NO NEW C ADDED TO $\mathcal{C}_{PROVEN}$

PROVEN CORE:

$$\begin{array}{c} T_* < \infty \\ \Downarrow \\ C_E^{weak} : \limsup \int_0^t W = +\infty \\ \wedge \\ C_{L^3} : \limsup \|u\|_3 = +\infty \end{array}$$

$$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}.$$

G-LAYER STATUS:

$$\boxed{C_G^{L^3} = \text{OPEN} : \|u(t_n)\|_{L^3(B_{r_n})} \geq \epsilon_0} \text{ from G-01/G-02 e7602f4.}$$

$$\boxed{C_G^{CKN} = \text{CONDITIONAL ON CKN-1982} : \mathcal{Q}(r_n) \geq \epsilon_{CKN}} \text{ from G-03 ffc8c30.}$$

$$\boxed{C_G^{geom} = \text{CANDIDATE/OPEN} : (\nabla \xi, L, \kappa, U)} \text{ from G-04 66ac917.}$$

$$\boxed{C_G^{L^3} \neq C_G^{CKN}} \text{ preserved, CROSS-E-L3-G-CKN-01 4f948bb COMPATIBLE/OPEN, no } \perp.$$

BOTTLENECK:

$$\boxed{C_E^{weak} \not\Rightarrow C_E^{strong}}$$

C_E^{strong} requires $W_- \in L^1(0, T_*)$, $W(t) = \int \omega^T S \omega = W_+ - W_-$.

C_E^{weak} gives $\int W_+ = \infty$ PROVEN, but not W_- control (E-strong-neg-02).

DIAGNOSTIC:

Hölder+Sobolev $\not\Rightarrow W_-$ control.

ESS-2003 $\Rightarrow C_{L^3}$ only, not W_- .

CKN-1982 $\Rightarrow C_G^{CKN}$ only, not W_- .

CF/DHY $\Rightarrow$ control of $|\alpha| = |\xi^T S \xi|$ or $\|\omega\|_\infty$, not $W_- \in L_t^1$.

Thus bottleneck is structural, not missing estimate.

NEXT DEEP QUESTION, NOT ANOTHER REGULARITY CRITERION:

What additional structure transforms $W_+ \rightarrow \infty$ into net control of W ?

Candidates: spectral sign structure of S , alignment $\omega \cdot e_i(S)$, coupling stretching-dissipation identity, not new norm.

G $\longrightarrow$ LOCKED. E-strong $\longrightarrow$ OPEN/BLOCKED as bottleneck. PHASE CLEAN.